Everyday Math

57. **Elevation** The highest elevation in the United States is Mount McKinley, Alaska, at $20{,}320$ feet above sea level. The lowest elevation is Death Valley, California, at 282 feet below sea level. Use integers to write the elevation of:
ⓐ Mount McKinley ⓑ Death Valley

58. **Extreme temperatures** The highest recorded temperature on Earth is $57°$ Celsius. The lowest recorded temperature is $90°$ below $0°$ Celsius. Use integers to write the:
ⓐ highest recorded temperature
ⓑ lowest recorded temperature

59. **State budgets** In June, 2011, the state of Pennsylvania estimated it would have a budget surplus of $\$540$ million. That same month, Texas estimated it would have a budget deficit of $\$27$ billion. Use integers to write the budget:
ⓐ surplus ⓑ deficit

60. **College enrollments** Across the United States, community college enrollment grew by $1{,}400{,}000$ students from 2007 to 2010. In California, community college enrollment declined by $110{,}171$ students from 2009 to 2010. Use integers to write the change in enrollment:
ⓐ growth ⓑ decline

Writing Exercises

61. Give an example of a negative number from your life experience.

62. What are the three uses of the "–" sign in algebra? Explain how they differ.

Self Check

ⓐ After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.

I can...	Confidently	With some help	No-I don't get it!
locate positive and negative numbers on the number line.			
order positive and negative numbers.			
find opposites.			
simplify expressions with absolute value.			
translate word phrases to expressions with integers.			

ⓑ If most of your checks were:

...confidently. Congratulations! You have achieved the objectives in this section. Reflect on the study skills you used so that you can continue to use them. What did you do to become confident of your ability to do these things? Be specific.

...with some help. This must be addressed quickly because topics you do not master become potholes in your road to success. In math, every topic builds upon previous work. It is important to make sure you have a strong foundation before you move on. Whom can you ask for help? Your fellow classmates and instructor are good resources. Is there a place on campus where math tutors are available? Can your study skills be improved?

...no—I don't get it! This is a warning sign and you must not ignore it. You should get help right away or you will quickly be overwhelmed. See your instructor as soon as you can to discuss your situation. Together you can come up with a plan to get you the help you need.

3.2 Add Integers

Learning Objectives

By the end of this section, you will be able to:

- Model addition of integers
- Simplify expressions with integers
- Evaluate variable expressions with integers
- Translate word phrases to algebraic expressions
- Add integers in applications

BE PREPARED 3.3 Before you get started, take this readiness quiz.

Evaluate $x + 8$ when $x = 6$.
If you missed this problem, review Example 2.13.

Simplify: $8 + 2(5 + 1)$.
If you missed this problem, review Example 2.8.

Translate *the sum of 3 and negative 7* into an algebraic expression.
If you missed this problem, review Table 2.7

Model Addition of Integers

Now that we have located positive and negative numbers on the number line, it is time to discuss arithmetic operations with integers.

Most students are comfortable with the addition and subtraction facts for positive numbers. But doing addition or subtraction with both positive and negative numbers may be more difficult. This difficulty relates to the way the brain learns.

The brain learns best by working with objects in the real world and then generalizing to abstract concepts. Toddlers learn quickly that if they have two cookies and their older brother steals one, they have only one left. This is a concrete example of $2 - 1$. Children learn their basic addition and subtraction facts from experiences in their everyday lives. Eventually, they know the number facts without relying on cookies.

Addition and subtraction of negative numbers have fewer real world examples that are meaningful to us. Math teachers have several different approaches, such as number lines, banking, temperatures, and so on, to make these concepts real.

We will model addition and subtraction of negatives with two color counters. We let a blue counter represent a positive and a red counter will represent a negative.



If we have one positive and one negative counter, the value of the pair is zero. They form a neutral pair. The value of this neutral pair is zero as summarized in Figure 3.17.

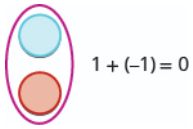


Figure 3.17 A blue counter represents $+1$. A red counter represents -1. Together they add to zero.

MANIPULATIVE MATHEMATICS

Doing the Manipulative Mathematics activity "Addition of signed Numbers" will help you develop a better understanding of adding integers.

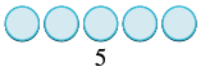
We will model four addition facts using the numbers $5, -5$ and $3, -3$.

$$5 + 3 \quad -5 + (-3) \quad -5 + 3 \quad 5 + (-3)$$

EXAMPLE 3.14

Model: $5 + 3$.

Solution

Interpret the expression.	$5 + 3$ means the sum of 5 and 3.
Model the first number. Start with 5 positives.	 5
Model the second number. Add 3 positives.	 5 3
Count the total number of counters.	 8 positives
The sum of 5 and 3 is 8.	$5 + 3 = 8$

TRY IT 3.27 Model the expression.

$2 + 4$

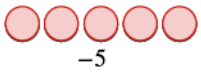
TRY IT 3.28 Model the expression.

$2 + 5$

EXAMPLE 3.15

Model: $-5 + (-3)$.

Solution

Interpret the expression.	$-5 + (-3)$ means the sum of -5 and -3.
Model the first number. Start with 5 negatives.	 −5
Model the second number. Add 3 negatives.	 −5 −3
Count the total number of counters.	 8 negatives
The sum of −5 and −3 is −8.	$-5 + -3 = -8$

TRY IT 3.29 Model the expression.

$-2 + (-4)$

TRY IT 3.30 Model the expression.

$-2 + (-5)$

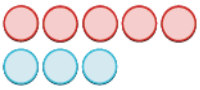
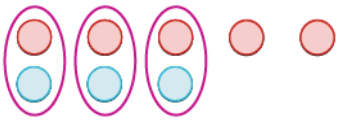
Example 3.14 and Example 3.15 are very similar. The first example adds 5 positives and 3 positives—both positives. The second example adds 5 negatives and 3 negatives—both negatives. In each case, we got a result of 8—either 8 positives or 8 negatives. When the signs are the same, the counters are all the same color.

Now let's see what happens when the signs are different.

EXAMPLE 3.16

Model: $-5 + 3$.

Solution

Interpret the expression.	$-5 + 3$ means the sum of -5 and 3.
Model the first number. Start with 5 negatives.	
Model the second number. Add 3 positives.	
Remove any neutral pairs.	
Count the result.	 2 negatives
The sum of −5 and 3 is −2.	$-5 + 3 = -2$

Notice that there were more negatives than positives, so the result is negative.

TRY IT 3.31 Model the expression, and then simplify:

$2 + (-4)$

TRY IT 3.32 Model the expression, and then simplify:

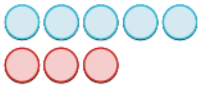
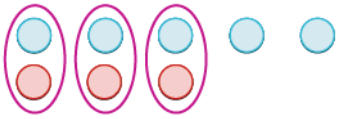
$2 + (-5)$

EXAMPLE 3.17

Model: $5 + (-3)$.

Solution

Interpret the expression.	$5 + (-3)$ means the sum of 5 and -3.
Model the first number. Start with 5 positives.	

Model the second number. Add 3 negatives.	
Remove any neutral pairs.	
Count the result.	 2 positives
The sum of 5 and −3 is 2.	$5 + (-3) = 2$

TRY IT 3.33 Model the expression, and then simplify:

$(-2) + 4$

TRY IT 3.34 Model the expression:

$(-2) + 5$

EXAMPLE 3.18

Modeling Addition of Positive and Negative Integers

Model each addition.

ⓐ 4 + 2 ⓑ −3 + 6 ⓒ 4 + (−5) ⓓ -2 + (−3)

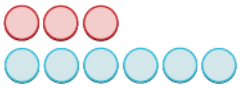
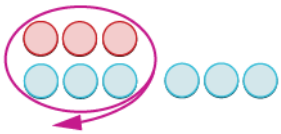
Solution

ⓐ

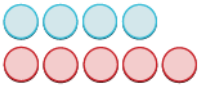
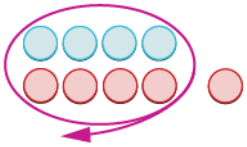
	$4 + 2$
Start with 4 positives.	
Add two positives.	
How many do you have?	$4 + 2 = 6$

ⓑ

	$-3 + 6$
Start with 3 negatives.	

Add 6 positives.	
Remove neutral pairs.	
How many are left?	
	3. $-3+6=3$

ⓒ

	$4+(-5)$
Start with 4 positives.	
Add 5 negatives.	
Remove neutral pairs.	
How many are left?	
	$-1.\ 4+(-5)=-1$

ⓓ

	$-2+(-3)$
Start with 2 negatives.	
Add 3 negatives.	
How many do you have?	$-5.\ -2+(-3)=-5$

TRY IT 3.35 Model each addition.

ⓐ 3 + 4 ⓑ −1 + 4 ⓒ 4 + (−6) ⓓ −2 + (−2)

TRY IT 3.36 ⓐ 5 + 1 ⓑ −3 + 7 ⓒ 2 + (−8) ⓓ −3 + (−4)

Simplify Expressions with Integers

Now that you have modeled adding small positive and negative integers, you can visualize the model in your mind to simplify expressions with any integers.

For example, if you want to add $37 + (-53)$, you don't have to count out 37 blue counters and 53 red counters.

Picture 37 blue counters with 53 red counters lined up underneath. Since there would be more negative counters than positive counters, the sum would be negative. Because $53-37 = 16$, there are 16 more negative counters.

$$37 + (-53) = -16$$

Let's try another one. We'll add $-74 + (-27)$. Imagine 74 red counters and 27 more red counters, so we have 101 red counters all together. This means the sum is -101.

$$-74 + (-27) = -101$$

Look again at the results of Example 3.14 - Example 3.17.

$5 + 3$	$-5 + (-3)$
both positive, sum positive	both negative, sum negative
When the signs are the same, the counters would be all the same color, so add them.	
$-5 + 3$	$5 + (-3)$
different signs, more negatives	different signs, more positives
Sum negative	sum positive
When the signs are different, some counters would make neutral pairs; subtract to see how many are left.	

Table 3.1 Addition of Positive and Negative Integers

EXAMPLE 3.19

Simplify:

ⓐ $19 + (-47)$ ⓑ $-32 + 40$

Solution

ⓐ Since the signs are different, we subtract 19 from 47. The answer will be negative because there are more negatives than positives.

$$19 + (-47)$$
$$-28$$

ⓑ The signs are different so we subtract 32 from 40. The answer will be positive because there are more positives than negatives

$$-32 + 40$$
$$8$$

TRY IT 3.37 Simplify each expression:

ⓐ $15 + (-32)$ ⓑ $-19 + 76$

TRY IT 3.38 Simplify each expression:

ⓐ $-55+9$ ⓑ $43+(-17)$

EXAMPLE 3.20

Simplify: $-14+(-36)$.

Solution

Since the signs are the same, we add. The answer will be negative because there are only negatives.

$$-14+(-36)$$
$$-50$$

TRY IT 3.39 Simplify the expression:

$-31+(-19)$

TRY IT 3.40 Simplify the expression:

$-42+(-28)$

The techniques we have used up to now extend to more complicated expressions. Remember to follow the order of operations.

EXAMPLE 3.21

Simplify: $-5+3(-2+7)$.

Solution

	$-5+3(-2+7)$
Simplify inside the parentheses.	$-5+3(5)$
Multiply.	$-5+15$
Add left to right.	10

TRY IT 3.41 Simplify the expression:

$-2+5(-4+7)$

TRY IT 3.42 Simplify the expression:

$-4+2(-3+5)$

Evaluate Variable Expressions with Integers

Remember that to evaluate an expression means to substitute a number for the variable in the expression. Now we can use negative numbers as well as positive numbers when evaluating expressions.

EXAMPLE 3.22

Evaluate $x+7$ when

ⓐ $x = -2$ ⓑ $x = -11$.

Solution

ⓐ Evaluate $x + 7$ when $x = -2$

	$-2 + 7$
Substitute -2 for x.	$-2 + 7$
Simplify.	5

ⓑ Evaluate $x + 7$ when $x = -11$

	$x + 7$
Substitute -11 for x.	$-11 + 7$
Simplify.	-4

TRY IT 3.43 Evaluate each expression for the given values:

$x + 5$ when

ⓐ $x = -3$ and ⓑ $x = -17$

TRY IT 3.44 Evaluate each expression for the given values: $y + 7$ when

ⓐ $y = -5$ ⓑ $y = -8$

EXAMPLE 3.23

When $n = -5$, evaluate

ⓐ $n + 1$ ⓑ $-n + 1$.

Solution

ⓐ Evaluate $n + 1$ when $n = -5$

	$n + 1$
Substitute -5 for n.	$-5 + 1$
Simplify.	-4

ⓑ Evaluate $-n + 1$ when $n = -5$

	$-n + 1$

Substitute −5 for n.	$-(-5)+1$
Simplify.	$5+1$
Add.	6

TRY IT 3.45 When $n=-8$, evaluate

(a) $n+2$ (b) $-n+2$

TRY IT 3.46 When $y=-9$, evaluate

(a) $y+8$ (b) $-y+8$.

Next we'll evaluate an expression with two variables.

EXAMPLE 3.24

Evaluate $3a+b$ when $a=12$ and $b=-30$.

Solution

	$3a+b$
Substitute 12 for a and −30 for b.	$3(12)+(-30)$
Multiply.	$36+(-30)$
Add.	6

TRY IT 3.47 Evaluate the expression:

$a+2b$ when $a=-19$ and $b=14$.

TRY IT 3.48 Evaluate the expression:

$5p+q$ when $p=4$ and $q=-7$.

EXAMPLE 3.25

Evaluate $(x+y)^2$ when $x=-18$ and $y=24$.

Solution

This expression has two variables. Substitute -18 for x and 24 for y.

	$(x+y)^2$
Substitute −18 for x and 24 for y.	$(-18+24)^2$

Add inside the parentheses.	$(6)^2$
Simplify	36

TRY IT 3.49 Evaluate:

$(x+y)^2$ when $x=-15$ and $y=29$.

TRY IT 3.50 Evaluate:

$(x+y)^3$ when $x=-8$ and $y=10$.

Translate Word Phrases to Algebraic Expressions

All our earlier work translating word phrases to algebra also applies to expressions that include both positive and negative numbers. Remember that the phrase *the sum* indicates addition.

EXAMPLE 3.26

Translate and simplify: the sum of -9 and 5.

Solution

The sum of −9 and 5 indicates addition.	the sum of -9 and 5
Translate.	$-9+5$
Simplify.	-4

TRY IT 3.51 Translate and simplify the expression:

the sum of -7 and 4

TRY IT 3.52 Translate and simplify the expression:

the sum of -8 and -6

EXAMPLE 3.27

Translate and simplify: the sum of 8 and -12, increased by 3.

Solution

The phrase *increased by* indicates addition.

	The sum of 8 and -12, increased by 3
Translate.	$[8+(-12)]+3$

Simplify.	$-4+3$
Add.	-1

TRY IT 3.53 Translate and simplify:

the sum of 9 and -16, increased by 4.

TRY IT 3.54 Translate and simplify:

the sum of -8 and -12, increased by 7.

Add Integers in Applications

Recall that we were introduced to some situations in everyday life that use positive and negative numbers, such as temperatures, banking, and sports. For example, a debt of $\$5$ could be represented as $-\$5$. Let's practice translating and solving a few applications.

Solving applications is easy if we have a plan. First, we determine what we are looking for. Then we write a phrase that gives the information to find it. We translate the phrase into math notation and then simplify to get the answer. Finally, we write a sentence to answer the question.

EXAMPLE 3.28

The temperature in Buffalo, NY, one morning started at 7 degrees below zero Fahrenheit. By noon, it had warmed up 12 degrees. What was the temperature at noon?

Solution

We are asked to find the temperature at noon.

Write a phrase for the temperature.	The temperature warmed up 12 degrees from 7 degrees below zero.
Translate to math notation.	−7 + 12
Simplify.	5
Write a sentence to answer the question.	The temperature at noon was 5 degrees Fahrenheit.

Table 3.2

TRY IT 3.55 The temperature in Chicago at 5 A.M. was 10 degrees below zero Celsius. Six hours later, it had warmed up 14 degrees Celsius. What is the temperature at 11 A.M.?

TRY IT 3.56 A scuba diver was swimming 16 feet below the surface and then dove down another 17 feet. What is her new depth?

EXAMPLE 3.29

A football team took possession of the football on their 42-yard line. In the next three plays, they lost 6 yards, gained 4 yards, and then lost 8 yards. On what yard line was the ball at the end of those three plays?

Solution

We are asked to find the yard line the ball was on at the end of three plays.

Write a word phrase for the position of the ball.	Start at 42, then lose 6, gain 4, lose 8.
Translate to math notation.	$42 - 6 + 4 - 8$
Simplify.	32
Write a sentence to answer the question.	At the end of the three plays, the ball is on the 32-yard line.

Table 3.3

TRY IT 3.57 The Bears took possession of the football on their 20-yard line. In the next three plays, they lost 9 yards, gained 7 yards, then lost 4 yards. On what yard line was the ball at the end of those three plays?

TRY IT 3.58 The Chargers began with the football on their 25-yard line. They gained 5 yards, lost 8 yards and then gained 15 yards on the next three plays. Where was the ball at the end of these plays?

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Adding Integers with Same Sign Using Color Counters (https://openstaxcollege.org/l/24samesigncount)
Adding Integers with Different Signs Using Counters (https://openstaxcollege.org/l/24diffsigncount)
Ex1: Adding Integers (https://openstaxcollege.org/l/24Ex1Add)
Ex2: Adding Integers (https://openstaxcollege.org/l/24Ex2Add)

SECTION 3.2 EXERCISES

Practice Makes Perfect

Model Addition of Integers

In the following exercises, model the expression to simplify.

63. $7 + 4$
64. $8 + 5$
65. $-6 + (-3)$
66. $-5 + (-5)$
67. $-7 + 5$
68. $-9 + 6$
69. $8 + (-7)$
70. $9 + (-4)$

Simplify Expressions with Integers

In the following exercises, simplify each expression.

71. $-21 + (-59)$
72. $-35 + (-47)$
73. $48 + (-16)$
74. $34 + (-19)$
75. $-200 + 65$
76. $-150 + 45$
77. $2 + (-8) + 6$
78. $4 + (-9) + 7$
79. $-14 + (-12) + 4$
80. $-17 + (-18) + 6$
81. $135 + (-110) + 83$
82. $140 + (-75) + 67$
83. $-32 + 24 + (-6) + 10$
84. $-38 + 27 + (-8) + 12$
85. $19 + 2(-3 + 8)$
86. $24 + 3(-5 + 9)$

Evaluate Variable Expressions with Integers

In the following exercises, evaluate each expression.

87. $x + 8$ when
ⓐ $x = -26$ ⓑ $x = -95$

88. $y + 9$ when
ⓐ $y = -29$ ⓑ $y = -84$

89. $y + (-14)$ when
ⓐ $y = -33$ ⓑ $y = 30$

90. $x + (-21)$ when
ⓐ $x = -27$ ⓑ $x = 44$

91. When $a = -7$, evaluate:
ⓐ $a + 3$ ⓑ $-a + 3$

92. When $b = -11$, evaluate:
ⓐ $b + 6$ ⓑ $-b + 6$

93. When $c = -9$, evaluate:
ⓐ $c + (-4)$
ⓑ $-c + (-4)$

94. When $d = -8$, evaluate:
ⓐ $d + (-9)$
ⓑ $-d + (-9)$

95. $m + n$ when, $m = -15$, $n = 7$

96. $p + q$ when, $p = -9$, $q = 17$

97. $r - 3s$ when, $r = 16$, $s = 2$

98. $2t + u$ when, $t = -6$, $u = -5$

99. $(a + b)^2$ when, $a = -7$, $b = 15$

100. $(c + d)^2$ when, $c = -5$, $d = 14$

101. $(x + y)^2$ when, $x = -3$, $y = 14$

102. $(y + z)^2$ when, $y = -3$, $z = 15$

Translate Word Phrases to Algebraic Expressions

In the following exercises, translate each phrase into an algebraic expression and then simplify.

103. The sum of -14 and 5

104. The sum of -22 and 9

105. 8 more than -2

106. 5 more than -1

107. -10 added to -15

108. -6 added to -20

109. 6 more than the sum of -1 and -12

110. 3 more than the sum of -2 and -8

111. the sum of 10 and -19, increased by 4

112. the sum of 12 and -15, increased by 1

Add Integers in Applications

In the following exercises, solve.

113. **Temperature** The temperature in St. Paul, Minnesota was $-19°\text{F}$ at sunrise. By noon the temperature had risen $26°\text{F}$. What was the temperature at noon?

114. **Temperature** The temperature in Chicago was $-15°\text{F}$ at 6 am. By afternoon the temperature had risen $28°\text{F}$. What was the afternoon temperature?

115. **Credit Cards** Lupe owes $\$73$ on her credit card. Then she charges $\$45$ more. What is the new balance?

116. **Credit Cards** Frank owes $\$212$ on his credit card. Then he charges $\$105$ more. What is the new balance?

117. **Football** A team lost 3 yards the first play. Then they lost 2 yards, gained 1 yard, and then lost 4 yards. What was the change in overall yardage over the four plays?

118. **Card Games** April lost 5 cards the first turn. Over the next three turns, she lost 3 cards, gained 2 cards, and then lost 1 card. What was the change in cards over the four turns?

119. **Football** The Rams took possession of the football on their own 35-yard line. In the next three plays, they lost 12 yards, gained 8 yards, then lost 6 yards. On what yard line was the ball at the end of those three plays?

120. **Football** The Cowboys began with the ball on their own 20-yard line. They gained 15 yards, lost 3 yards and then gained 6 yards on the next three plays. Where was the ball at the end of these plays?

121. **Scuba Diving** A scuba diver swimming 8 feet below the surface dove 17 feet deeper; the pressure got to them and they rose five feet. What is their new depth?

122. **Gas Consumption:** Ozzie rode their motorcycle for 30 minutes, using 168 fluid ounces of gas. Then they stopped and got 140-fluid ounces of gas. Represent the change in gas amount as an integer.

Everyday Math

123. **Stock Market** The week of September 15, 2008, was one of the most volatile weeks ever for the U.S. stock market. The change in the Dow Jones Industrial Average each day was:

Monday	−504	Tuesday	+142	Wednesday	−449
Thursday	+410	Friday	+369		

What was the overall change for the week?

124. **Stock Market** During the week of June 22, 2009, the change in the Dow Jones Industrial Average each day was:

Monday	−201	Tuesday	−16	Wednesday	−23
Thursday	+172	Friday	−34		

What was the overall change for the week?

Writing Exercises

125. Explain why the sum of −8 and 2 is negative, but the sum of 8 and −2 and is positive.

126. Give an example from your life experience of adding two negative numbers.

Self Check

ⓐ After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.

I can...	Confidently	With some help	No-I don't get it!
model addition of integers.			
simplify expressions with integers.			
evaluate variable expressions with integers.			
translate word phrases to algebraic expressions.			
add integers in applications.			